MOTILAL OSWAL

S&P 500 Power BI Dashboard

Complete Replication Guide

DAX Measures | Card Formulas | Chart Mappings | Data Columns

Dataset: S&P 500 Stock Prices 2014-2017 | Columns: symbol, date, open, high, low, close, volume

SECTION 0: DATASET & TABLE SETUP

Your Raw Dataset Columns

The CSV file contains exactly 7 columns. Every DAX measure and chart in this dashboard is built from these columns only:

Column Name	Data Type	Description
symbol	Text	Stock ticker (e.g., AAPL, MSFT, NVDA) — used as the primary slicer/filter
date	Date	Trading date in DD-MM-YYYY format — MUST be converted to Date type in Power Query
open	Decimal Number	Opening price of the stock for that trading day
high	Decimal Number	Highest price reached during that trading day
low	Decimal Number	Lowest price reached during that trading day
close	Decimal Number	Closing price — MOST USED column across all charts and measures
volume	Whole Number	Number of shares traded — used for volume charts and AVG Volume measure

Power Query Setup (Do This First)

- Go to Transform Data > Power Query Editor
- Select the 'date' column > Change Type > Date
- Add a 'Year' column: Add Column > Date > Year
- Add a 'Month' column: Add Column > Date > Month
- Add a 'MonthName' column: Add Column > Date > Name of Month
- Add a 'Quarter' column: Add Column > Date > Quarter of Year

NOTE: The date column format is DD-MM-YYYY. Power BI may auto-detect it wrongly. Manually set the locale to match during import.

SECTION 1: ALL DAX MEASURES — FORMULAS & EXPLANATIONS

Create all these measures by going to: Home > New Measure (in the Report view). Organise them inside a 'Measure Table' by creating a blank table named 'Measures'.

Measure 1: AVG Volumn

NOTE: This is the measure already in your dashboard labelled 'AVG Volumn' (keep that exact name for compatibility).

```
AVG Volumn =
 AVERAGE( StockPrices[volume] )
```

Explanation:

AVERAGE() calculates the mean of all values in the [volume] column, automatically respecting any active filters (symbol slicer, date slicer, etc.). When you select AAPL on your slicer, this returns the average daily trading volume for AAPL only.

Measure 2: MA 30 Days

NOTE: Moving Average over the last 30 calendar days of close price.

```
MA 30 Days =
 CALCULATE(
     AVERAGE( StockPrices[close] ),
     DATESINPERIOD(
         StockPrices[date],
         LASTDATE( StockPrices[date] ),
         -30,
         DAY
     )
 )
```

Explanation:

DATESINPERIOD() creates a dynamic window of dates: starting from the last date in the current context, going back 30 days. CALCULATE() then evaluates AVERAGE(close) only within those 30 days. On a line chart with date on the X-axis, each point shows the 30-day rolling average up to that date — this smooths out short-term price noise.

Measure 3: Return YTD

NOTE: Year-to-date return from Jan 1st of the current year to the latest date in context.

```
Return YTD =
 VAR StartPrice =
     CALCULATE(
         AVERAGE( StockPrices[close] ),
         DATESMTD( STARTOFYEAR( StockPrices[date] ) )
     )
 VAR EndPrice =
     CALCULATE(
         AVERAGE( StockPrices[close] ),
         LASTDATE( StockPrices[date] )
     )
 RETURN
     DIVIDE( EndPrice - StartPrice, StartPrice ) * 100
```

Explanation:

VAR creates reusable variables inside the DAX formula. StartPrice finds the average close at the beginning of the year. EndPrice finds the most recent average close. DIVIDE() safely handles division by zero. Multiply by 100 converts to percentage. Result: '+19.4%' style output.

Measure 4: Total Return %

NOTE: Total return from the very first date in the dataset (Jan 2014) to the last date in context.

```
Total Return % =
 VAR FirstDate =
     CALCULATE(
         AVERAGE( StockPrices[close] ),
         FIRSTDATE( ALL( StockPrices[date] ) )
     )
 VAR LastDate =
     CALCULATE(
         AVERAGE( StockPrices[close] ),
         LASTDATE( StockPrices[date] )
     )
 RETURN
     DIVIDE( LastDate - FirstDate, FirstDate ) * 100
```

Explanation:

ALL(StockPrices[date]) removes any date filter so FIRSTDATE always returns the absolute earliest date in the table (Jan 2014). This gives the full multi-year return percentage. When AAPL is selected on the slicer, this returns AAPL's total return from 2014 to 2017.

Measure 5: Stock Rank

NOTE: Ranks each stock by Total Return % — rank 1 = highest return.

```
Stock Rank =
 RANKX(
     ALL( StockPrices[symbol] ),
     [Total Return %],
     ,
     DESC,
     DENSE
 )
```

Explanation:

RANKX() compares the current symbol's Total Return % against all other symbols (ALL removes the symbol filter). DESC means rank 1 = best performer. DENSE means no gaps in ranking if two stocks tie. Used in the KPI card 'Rank 47/505' on Page 2.

Measure 6: Top Return %

NOTE: Returns the single highest Total Return % across all stocks.

```
Top Return % =
 MAXX(
     ALL( StockPrices[symbol] ),
     [Total Return %]
 )
```

Explanation:

MAXX() iterates over every symbol (ALL removes filters) and returns the maximum value of [Total Return %]. This gives you the '+312%' figure shown on the Executive Overview KPI card. It does NOT change when you use a symbol slicer since ALL() ignores it.

Measure 7: Top Stock

NOTE: Returns the symbol name of the best-performing stock.

```
Top Stock =
 CALCULATE(
     SELECTEDVALUE( StockPrices[symbol] ),
     TOPN(
         1,
         ALL( StockPrices[symbol] ),
         [Total Return %],
         DESC
     )
 )
```

Explanation:

TOPN(1, ...) finds the single row with the highest Total Return %. SELECTEDVALUE() then reads the symbol from that row. Returns text like 'NVDA'. Used in card subtitles like 'Best performer: NVDA'.

Measure 8: Rank Top

NOTE: Count of stocks in the Top 10 return bracket — used for the rank distribution bar chart.

```
Rank Top =
 CALCULATE(
     COUNTROWS( VALUES( StockPrices[symbol] ) ),
     FILTER(
         ALL( StockPrices[symbol] ),
         [Stock Rank] <= 10
     )
 )
```

Explanation:

Filters the symbol table to only those where Stock Rank is 10 or better, then counts them. Useful as a card showing '10 stocks in Top Rank'. Modify the number (<=10, <=25, etc.) to create bracket variants.

Measure 9: Rank Worst

NOTE: The worst rank number in the current filter context — i.e., total number of stocks being ranked.

```
Rank Worst =
 MAXX(
     ALL( StockPrices[symbol] ),
     [Stock Rank]
 )
```

Explanation:

Returns the maximum rank number (e.g., 505), confirming how many stocks are in the full dataset. Used to show 'Rank X of 505' style labels.

Measure 10: Volatility %

NOTE: Annualised price volatility — standard measure of risk in finance.

```
Volatility % =
 VAR DailyReturns =
     ADDCOLUMNS(
         StockPrices,
         "DailyRet",
         DIVIDE(
             StockPrices[close] - StockPrices[open],
             StockPrices[open]
         )
     )
 VAR StdDev =
     STDEVX.P( DailyReturns, [DailyRet] )
```

```
RETURN
    StdDev * SQRT( 252 ) * 100
```

Explanation:

Daily return = (Close - Open) / Open for each row. STDEVX.P calculates the population standard deviation of all daily returns. Multiply by SQRT(252) to annualise (252 = trading days in a year). Multiply by 100 for percentage. Higher = riskier stock.

SECTION 2: ADDITIONAL RECOMMENDED DAX MEASURES

These 6 measures are NOT in your current list but are recommended to make the dashboard finance-grade and stand out on your resume.

Measure 11: Sharpe Ratio (Approx)

```
Sharpe Ratio =
 DIVIDE( [Return YTD], [Volatility %] )
```

Risk-adjusted return. A Sharpe > 1 is considered good. Shows quality of returns relative to risk taken.

Measure 12: Price Range %

```
Price Range % =
 DIVIDE(
     AVERAGE( StockPrices[high] ) - AVERAGE( StockPrices[low] ),
     AVERAGE( StockPrices[close] )
 ) * 100
```

Intraday price movement as a percentage of close price. High values = more volatile intraday behaviour.

Measure 13: Volume Surge Flag

```
Volume Surge Flag =
 IF(
     AVERAGE( StockPrices[volume] ) > 2 * [AVG Volumn],
     "SURGE",
     "Normal"
 )
```

Detects when current-context average volume is more than 2x the baseline average. Use as a slicer or conditional icon in tables.

NOTE: For this to work correctly, the first [AVG Volumn] should be calculated over a fixed baseline (e.g. full year) using CALCULATE with ALL(date) to avoid circular reference.

Measure 14: Drawdown %

```
Drawdown % =
 VAR MaxClose =
     CALCULATE(
         MAX( StockPrices[close] ),
         ALLSELECTED( StockPrices[date] )
     )
 VAR CurrentClose =
     AVERAGE( StockPrices[close] )
 RETURN
     DIVIDE( CurrentClose - MaxClose, MaxClose ) * 100
```

Shows how far the current price is from its peak within the selected period. A drawdown of -30% means price is 30% below its highest point. Risk metric.

Measure 15: Momentum Score

```
Momentum Score =
 VAR Return3M =
     CALCULATE(
         [Return YTD],
         DATESINPERIOD( StockPrices[date], LASTDATE( StockPrices[date] ), -90, DAY )
     )
 VAR Return6M =
     CALCULATE(
         [Return YTD],
         DATESINPERIOD( StockPrices[date], LASTDATE( StockPrices[date] ), -180, DAY )
     )
 RETURN
     DIVIDE( Return3M, ABS( Return6M ) + 0.001 )
```

Ratio of 3-month return to 6-month return. Score > 1 means stock is accelerating (recent returns outpacing longer trend). Used for momentum-based filtering.

Measure 16: YoY Return %

```
YoY Return % =
 VAR CurrentYearReturn =
     CALCULATE(
         [Total Return %],
         DATESYTD( StockPrices[date] )
     )
 VAR PriorYearReturn =
     CALCULATE(
         [Total Return %],
         DATESYTD( DATEADD( StockPrices[date], -1, YEAR ) )
     )
 RETURN
     CurrentYearReturn - PriorYearReturn
```

Year-over-Year change in return %. This drives the card label 'YoY +22.1%' on Page 1. Positive = improving performance vs prior year.

SECTION 3: KPI CARD FORMULAS — ALL 3 PAGES

PAGE 1 — Executive Overview KPI Cards

Card Title	Main Value (DAX)	Subtitle Label	Subtitle DAX
Total Return %	[Total Return %]	YoY +22.1%	[YoY Return %] formatted as +0.0%;-0.0%
Top Return %	[Top Return %]	Best: 2017	"Best: " & [Top Stock Year] *
Avg Daily Volume	[AVG Volumn]	▲ 8.3% vs prior yr	DIVIDE([AVG Volumn]-[AVG Volumn PY],[AVG Volumn PY])*100 **
Volatility %	[Volatility %]	▲ Peak 2015	"Peak: " & [Peak Vol Year] *

Card Title	Main Value (DAX)	Subtitle Label	Subtitle DAX
Return YTD 2017	[Return YTD]	3yr CAGR 14%	[CAGR 3Y] ***

*NOTE: * Top Stock Year and Peak Vol Year are helper text measures using TOPN + CALCULATE to find the year with max return or max volatility. See helper measures below.*

*NOTE: ** AVG Volumn PY = CALCULATE([AVG Volumn], SAMEPERIODLASTYEAR(StockPrices[date]))*

*NOTE: *** CAGR 3Y = (POWER(1 + [Total Return %]/100, 1/3) - 1) * 100*

Helper Measures for Card Subtitles

```
-- Year of Top Return
 Top Stock Year =
 CALCULATE(
     SELECTEDVALUE( StockPrices[Year] ),
     TOPN( 1, ALL( StockPrices[Year] ), [Total Return %], DESC )
 )

 -- Year of Highest Volatility
 Peak Vol Year =
 CALCULATE(
     SELECTEDVALUE( StockPrices[Year] ),
     TOPN( 1, ALL( StockPrices[Year] ), [Volatility %], DESC )
 )

 -- 3-Year CAGR
 CAGR 3Y =
 ( POWER( 1 + [Total Return %] / 100, 1/3 ) - 1 ) * 100

 -- Prior Year Avg Volume
 AVG Volumn PY =
 CALCULATE(
     [AVG Volumn],
     SAMEPERIODLASTYEAR( StockPrices[date] )
 )
```

PAGE 2 — Stock Deep Dive KPI Cards

Card Title	Main Value (DAX)	Subtitle Label	Subtitle DAX
Total Return %	[Total Return %]	Rank 47/505	[Stock Rank] & "/" & [Rank Worst]
MA 30 Days	[MA 30 Days]	Above/Below 30d avg	IF([close]>[MA 30 Days],"Above 30d avg","Below 30d avg")
Avg Volume	[AVG Volumn]	High/Low liquidity	IF([AVG Volumn]>10000000,"High liquidity","Low liquidity")
Volatility %	[Volatility %]	Above/Below avg	IF([Volatility %]>[Avg Mkt Vol],"Above avg","Below avg")

NOTE: For the MA 30 Days card: [close] here means LASTNONBLANK(StockPrices[close],1) to get the most recent close price as a scalar.

```
-- Latest Close Price (scalar for card comparisons)
 Latest Close =
```

```
CALCULATE(
    AVERAGE( StockPrices[close] ),
    LASTDATE( StockPrices[date] )
)

-- Market Average Volatility (for comparison)
Avg Mkt Vol =
CALCULATE(
    [Volatility %],
    ALL( StockPrices[symbol] )
)
```

PAGE 3 — Risk & Portfolio KPI Elements

Page 3 uses visual KPI tiles (not standard cards). Each stock's volatility tile is built using a Matrix visual or multi-row card:

Visual Element	Measure Used	Conditional Formatting Rule
NFLX 34%	[Volatility %] filtered to NFLX	Background Red (#E74C3C) if value > 30
NVDA 28%	[Volatility %] filtered to NVDA	Background Orange (#E07B20) if 20-30
AMZN 22%	[Volatility %] filtered to AMZN	Background Gold (#F5A623) if 20-25
AAPL 18%	[Volatility %] filtered to AAPL	Background Navy (#1E2B8E) if 15-20
GOOG 17%	[Volatility %] filtered to GOOG	Background Green (#27AE60) if < 20
MSFT 15%	[Volatility %] filtered to MSFT	Background Green (#27AE60) if < 20

NOTE: In Power BI: use a Multi-Row Card visual. Set the field to [Volatility %], apply a Symbol slicer to filter, then apply conditional background color formatting using the rules above.

SECTION 4: CHART DATA MAPPINGS — ALL 3 PAGES

Each chart entry below specifies the exact Power BI visual type, which column or measure goes on which axis/field, and any special formatting notes.

PAGE 1 — Executive Overview Chart Mappings

Chart 1: Line Chart — "Avg Close Price Trend + MA 30 Days"

X Axis: StockPrices[date] (set to Month hierarchy: Year > Quarter > Month)
Y Axis (Primary): [Total Return %] → rename to 'Avg Close'; actually use AVERAGE(close) or create measure: Avg Close Price = AVERAGE(StockPrices[close])
Secondary Line: [MA 30 Days] — add as second Y value; format as dashed gold line
Legend: None (two series auto-legend)
Filters: Page-level: Year slicer (2014-2017)

Chart 2: Clustered Column Chart — "Total Return % by Year"

X Axis: StockPrices[Year] (the column you created in Power Query)
Y Axis: [Total Return %]

Color: Conditional formatting: positive values = Navy (#1E2B8E), negative = Red (#E74C3C)
Data Labels: On — format as +0.0%;-0.0%
Sort: X axis sorted by Year ascending

Chart 3: Combo Chart (Bar + Line) — "Volume vs Price Monthly"

X Axis (Shared): StockPrices[date] at Month level (MonthName or Month number)
Column Y Axis: [AVG Volumn] — displayed as bars (navy fill)
Line Y Axis: Avg Close Price = AVERAGE(StockPrices[close]) — gold line, secondary axis
Secondary Axis: Enable: turn on secondary Y axis for the line series
Slicer: Symbol slicer filters both series simultaneously

Chart 4: Donut Chart — "Top Sectors by Avg Return"

Legend (Slices): Sector — NOTE: your dataset has no Sector column. Create a calculated column using SWITCH: Sector = SWITCH(StockPrices[symbol], "AAPL","Tech", "MSFT","Tech", "AMZN","Tech", "GS","Finance", ... "Other")
Values: [Total Return %] — aggregated per sector
Colors: Custom: Tech=Navy, Finance=Gold, Health=Green, Energy=Purple, Consumer=Red
Detail Labels: Category + Percentage

Chart 5: Horizontal Bar Chart (Rank Buckets) — "Return by Rank Bracket"

Y Axis (Categories): Create a static table: Rank Buckets = DATATABLE("Bucket", STRING, {{"Top 10"},{"Top 25"},{"Mid 50"},{"Bot 25"},{"Bot 10"}})
X Axis (Values): [Total Return %] filtered per bucket using CALCULATE + FILTER on [Stock Rank]
Colors: Conditional: Top=Navy gradient, Bottom=Red gradient
Data Labels: On — show return % at end of bar

PAGE 2 — Stock Deep Dive Chart Mappings

Chart 1: Line Chart (simulating Candlestick) — "OHLC Price Movement"

X Axis: StockPrices[date] — daily level (no hierarchy grouping)
Y Axis — Line 1 (Close): AVERAGE(StockPrices[close]) — navy, 1.5px line
Y Axis — Line 2 (High): AVERAGE(StockPrices[high]) — green, 1px dashed line
Y Axis — Line 3 (Low): AVERAGE(StockPrices[low]) — red, 1px dashed line
True Candlestick Option: Install 'Candlestick Chart' by OKViz from AppSource. Fields: Date, Open=open, High=high, Low=low, Close=close
Slicer Dependency: Symbol slicer — shows one stock at a time for meaningful OHLC

Chart 2: Combo Chart — "Daily Volume with Avg Line"

X Axis: StockPrices[date] at daily level
Bar Y Axis: StockPrices[volume] — SUM or individual values (navy bars)
Line Y Axis: [AVG Volumn] — constant reference line (gold dashed)
Tooltip: Add: % diff from avg = DIVIDE(SUM(volume)-[AVG Volumn],[AVG Volumn])*100
Note: Set bar chart to NOT aggregate — use individual daily values for volume

Chart 3: Scatter Chart — "Risk-Return Scatter"

X Axis: [Volatility %] — risk dimension
Y Axis: [Total Return %] — return dimension
Size (Bubble): [AVG Volumn] — larger bubble = higher trading volume
Legend (Color): Sector calculated column (see Chart 4 Page 1 for Sector formula)
Details: StockPrices[symbol] — so each bubble represents one stock

Reference Lines: Add average lines: X = AVERAGE([Volatility %]), Y = AVERAGE([Total Return %]) to create 4 quadrants

Chart 4: Area Chart — "Stock Rank Over Time"

X Axis: StockPrices[date] at Month level
Y Axis: [Stock Rank] — lower = better; consider inverting axis
Fill: Navy with 10% transparency
Drillthrough: Enable drillthrough: right-click on data point > Drill through > Page 2
Axis Invert: Consider inverting Y-axis so rank 1 appears at top (Format > Y Axis > Invert range)

Chart 5: Matrix / Table — "Top 10 Stocks Table"

Rows: StockPrices[symbol]
Column 1 — Rank: [Stock Rank]
Column 2 — Total Return %: [Total Return %] — apply data bar conditional formatting (navy)
Column 3 — Avg Volume: [AVG Volumn]
Column 4 — Volatility %: [Volatility %] — apply icon set: green circle <20%, orange triangle 20-30%, red diamond >30%
Column 5 — Return YTD: [Return YTD]
Sparklines: Right-click column header > Add sparkline > choose [close] over date
Top N Filter: Add visual-level filter: Stock Rank <= 10

PAGE 3 — Risk & Portfolio Chart Mappings

Chart 1: Scatter Chart — "Risk-Return Quadrant Map"

X Axis: [Volatility %] — one value per symbol
Y Axis: [Total Return %] — one value per symbol
Details: StockPrices[symbol] — each dot = one stock
Color Saturation: [Total Return %] — use diverging color scale: navy (high) to red (negative)
Quadrant Lines: Add constant line on X axis at AVERAGEX(ALL(symbol),[Volatility %]); same for Y axis
Quadrant Labels: Add text boxes manually: 'Star' (high return, low vol), 'High Risk' (high return, high vol), 'Value' (low return, low vol), 'Avoid' (low return, high vol)

Chart 2: Multi-Row Card or Gauge — "Volatility % Top 6 Stocks"

Field: [Volatility %]
Filter: Apply visual-level filter: symbol IN (NFLX, NVDA, AMZN, AAPL, GOOG, MSFT)
Conditional Background: Format > Data labels > Background color > Rules: >30=Red, 20-30=Orange, <20=Green
Display: Format as percentage with 0 decimal places: 0%

Chart 3: Line Chart (Dual Series) — "MA 30 Days vs Close Price"

X Axis: StockPrices[date] at Month level
Y Axis — Line 1 (Close): Avg Close Price = AVERAGE(StockPrices[close]) — navy, thin 1px
Y Axis — Line 2 (MA): [MA 30 Days] — gold, bold 2px
Area Fill: Enable 'Fill below line' for one series to shade the gap
Legend: Both series labelled: 'Close Price' and 'MA 30 Days'
Slicer Dependency: Symbol slicer — meaningful only for individual stocks

Chart 4: Matrix (Heatmap) — "Monthly Return Heatmap"

Rows: StockPrices[Year]
Columns: StockPrices[MonthName] — sort by Month number (1-12) not alphabetically
Values: [Total Return %] — calculated at the Year+Month intersection

Conditional Formatting: Background color: 5-stop diverging scale. Min=Red (#E74C3C), Center=Gray (#B4B2A9), Max=Navy (#1E2B8E). Apply to cell background.
Font Color: White for all cells (for contrast against colored backgrounds)
Month Sort Fix: In the MonthName column: Modeling > Sort by Column > Month (number)

SECTION 5: STEP-BY-STEP BUILD ORDER

Follow this exact sequence in Power BI Desktop:

Step	Task	Details
1	Import CSV	Get Data > Text/CSV > select file. Set date column locale correctly.
2	Power Query transforms	Change date type, add Year, Month, MonthName, Quarter columns. Close & Apply.
3	Create Sector column	New calculated column using SWITCH on symbol for donut chart grouping.
4	Create Measure Table	Enter Data > blank table named 'Measures'. Hides clutter from data table.
5	Build all 10 core measures	In order: AVG Volumn > MA 30 Days > Total Return % > Return YTD > Stock Rank > Top Return % > Top Stock > Rank Top > Rank Worst > Volatility %
6	Build helper measures	YoY Return %, CAGR 3Y, AVG Volumn PY, Latest Close, Avg Mkt Vol, Top Stock Year, Peak Vol Year
7	Build recommended measures	Sharpe Ratio, Price Range %, Volume Surge Flag, Drawdown %, Momentum Score
8	Build Page 1	Add header, 5 KPI cards, then all charts. Set page-level date slicer.
9	Build Page 2	Add header, symbol+year slicers, 4 KPI cards, all charts, matrix table.
10	Build Page 3	Add header, scatter quadrant, vol tiles, MA line chart, heatmap matrix.
11	Enable Drillthrough	Page 2: Drillthrough > add symbol field. Page 1 scatter: right-click > Drill through.
12	Format & Theme	Set theme colors to Motilal Oswal palette. Navy=#1E2B8E, Gold=#F5A623.

NOTE: Always build measures BEFORE building visuals. Visuals depend on measures, not the other way around.